Adaptive Prevention of Computer Vision Syndrome with Edge-Based Camera-Only Smart Glasses

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***Abstract*—The pervasive use of digital screens has catalyzed a global rise in Computer Vision Syndrome (CVS). Current wearable assistive technologies often rely on bulky multi-sensor arrays or cloud-based processing, compromising user ergonomics and privacy. This paper presents a camera-only smart glasses system engineered to autonomously mitigate CVS by operationalizing the 20-20-20 rule. Rather than functioning as a static timer, the system operates as a dynamic, adaptive mechanism that adjusts in real time to user behavior: a front facing micro-camera and an edge computing processor running a local AI agent analyze the egocentric visual stream to discriminate active, light-emitting screen use from non-digital near-work, and trigger unobtrusive haptic micro-vibrations once an unbroken screen viewing threshold is accumulated. To support robust on-device inference, we construct a custom egocentric dataset and incorporate confounding elements, such as bright windows and reflective surfaces, into the screen-absent class, extending the classifier's decision boundary to real world conditions. The edge-optimized model reliably distinguishes screen-active from screen-absent states across diverse screen types and lighting conditions while operating within the memory and power constraints of a wearable microcontroller. This work demonstrates that embedding complex vision tasks directly onto wearable edge devices is feasible, offering a privacy preserving, low power solution for CVS mitigation that is scalable to diverse digital platforms.**

***Index Terms*—Computer Vision Syndrome, smart glasses, edge computing, 20-20-20 rule, MobileNet, haptic feedback, egocentric vision**

I. Introduction

The continuous progress of wearable technologies and Artificial Intelligence (AI) is radically changing how we interact with our surroundings. Smart wearable devices, such as smartwatches, smart glasses, and biometric sensors, are becoming increasingly widespread. Thanks to the miniaturization of microelectronic components and increased processing capabilities, these devices are no longer mere accessories but advanced computing tools capable of passively monitoring and supporting human activities in real time. Their diffusion has also grown significantly in recent years: a large-scale survey conducted in the United States between 2020 and 2022 found that nearly half of respondents reported owning a wearable device, highlighting how these technologies are becoming progressively integrated into everyday life and health-related practices [1]. The seamless integration of wearable healthcare devices into high-cognitive-demand everyday settings enables user-centric applications that extend beyond physical health monitoring. These devices encourage adoption for healthier lifestyles, facilitating lasting behavioral changes in daily habits [2].

The rapid and pervasive integration of digital technology into modern society has precipitated a profound environmental and evolutionary mismatch, introducing unprecedented challenges related to visual and cognitive fragmentation. In contemporary occupational and educational environments, the concept of "continuous partial attention" dictates that users frequently fragment their focus across multiple light-emitting devices simultaneously. Rather than merely a crisis of attention, this continuous flow of digital stimuli has catalyzed a global pandemic of Computer Vision Syndrome (CVS), with recent meta-analyses establishing a pooled global prevalence of 66 percent among device users, surging as high as 94.5 percent among university cohorts during periods of intense digital reliance [3]. This chronic overstimulation and uninterrupted near-vision work exacts a severe physiological and cognitive toll (see Fig. 1). The immense cognitive demand required during active digital interaction severely suppresses autonomic functions such as the blink reflex leading to profound visual exhaustion, tension headaches, and degraded sleep architecture [3] [4] [5]. Ultimately, this fragmentation and the resulting physical strain directly diminish cognitive endurance, contributing to a staggering global economic burden in lost workforce productivity.

Current research on wearable technologies increasingly emphasizes continuous and objective sensing to support health-related behaviors in daily life [3]. Rather than relying on static software-based timers, modern systems integrate advanced sensing capabilities directly into smart eyewear, enabling context-aware detection of user activity and timely interventions [6]. Furthermore, instead of using intrusive webcam-based eye-tracking, cutting-edge edge-computing models integrate spatial sensors (LiDAR) to measure exact viewing distances [7] and innovative low power radio wave sensing to track intricate blink dynamics such as blink duration and physiological completeness [6]. This allows the eyewear to autonomously detect the eye dryness and verify true biological compliance with restorative protocols like the 20-20-20 rule [6] [7].

This work focuses on a smart-glasses-based assistive system explicitly engineered to mitigate Computer Vision Syndrome (CVS) during prolonged digital engagement across educational and occupational settings [3]. Rather than utilizing embed-

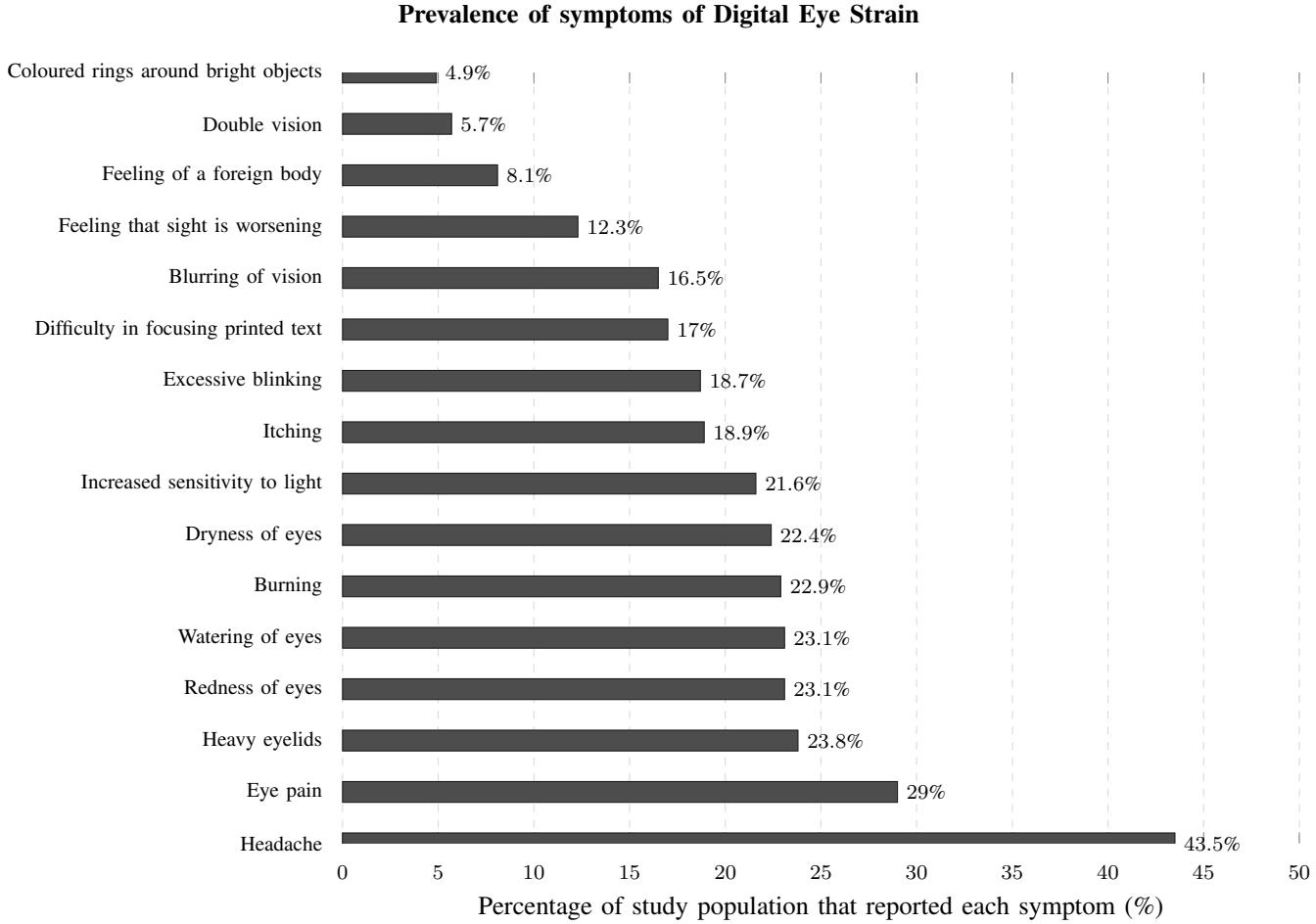


Fig. 1. Graph depicting the frequency with which each symptom of digital eye strain was experienced by the study population.

ded cameras to invasively track cognitive attention or eye movements, our approach strictly leverages a forward-facing camera dedicated solely to detecting digital screens and light-emitting displays in the user's environment [8]. By analyzing this continuous visual stream, the system objectively monitors cumulative screen exposure without interfering with normal binocular vision [9]. Building on these camera-based inputs, we utilize advanced edge-computing deep learning algorithms to seamlessly discriminate between active screen use and general non-digital near-work [10]. When an unbroken 20-minute threshold of screen viewing is accumulated, the system delivers real-time, unobtrusive feedback via peripheral hardware cues, such as subtle haptic vibrations. This intervention gently prompts the user to avert their gaze to optical infinity (20 feet) for 20 seconds, thereby actively operationalizing the clinically endorsed 20-20-20 rule to restore tear film stability, release intraocular muscular tension, and safeguard long-term visual health [11] [12].

The primary contribution of this research is the development of a camera-only smart glasses system engineered to autonomously detect digital screen exposure in real time. By relying strictly on the visual stream from an embedded forward-facing camera, the system successfully distinguishes active, light-emitting screen use from non-digital near-work (such as reading physical textbooks or writing) to combat the onset of Computer Vision Syndrome (CVS).

II. Related works

The growing pervasiveness of digital screens and the associated surge in Computer Vision Syndrome (CVS) have spurred a broad line of research in wearable sensing, computer vision, and human–computer interaction. This section maps the main research axes directly relevant to our smart-glasses-based system, highlighting both the strengths and limitations of existing approaches and situating our contribution within this landscape.

A. Heuristic Software Interventions and Static Systems

The very first generation of systems is based on local applications or web extensions that act as timers, such as "Eye Care 20 20 20" and "Eye Rest Notification" [13] [14]. The fundamental merit lies in the absence of additional hardware, guaranteeing low-cost implementation and the absolute maintenance of user privacy. These software are "blind" with respect to the real visual context: they do not distinguish focal

attention from simple mouse inactivity. These generate full-screen notifications that brutally interrupt the cognitive load, leading to habituation. Being confined to the single host, they do not calculate cumulative exposure across multiple devices [15] [12].

B. Ambient Vision and Non-Wearable Spatial Estimation

To acquire contextual awareness without on-body sensors, external webcams are used for postural analysis and facial distance measurement. In parallel, monocular neural networks estimate spatial depth in real time. The main advantage is that these systems are able to obtain objective postural metrics, effectively filtering out temporary false alarms related to physiological micro-movements. However, these approaches suffer from critical spatial constraints (location-bound) that prevent tracking while on the move and they are embedded devices that require significant GPU resources. Finally, constantly pointing external cameras at workers raises unacceptable privacy concerns [16] [17].

C. Wearable Photometric Sensors (Smart Glasses)

By transferring sensors onto the user, some assistive glasses exploit color (RGB) light micro-sensors to monitor the specific irradiance of blue light (464 nm) emitted by LEDs. Their positioning ensures the exact tracking of corneal exposure cross-device. They consume very little energy and intrinsically protect privacy, since they do not process video streams. They present functional blindness in very bright natural scenarios, where the screen light disappears. Furthermore, natural head movements alter photometric streams, severely limiting their daily reliability [7] [18].

D. Multi-Sensory Fusion Wearable Architectures

To stabilize the optical sensors, architectures like "Tiger" resort to a multi-sensory array that combines RGB sensors, Inertial Units (IMU) for head stability, and LiDAR for screen distance. It has superb precision in validating moments of true visual pause with the compromise that necessitates a bulky hardware. Integrating three types of sensors causes excessive thickness, overheating, and operational consumption totally incompatible with daily and invisible use [19] [20].

E. Mechanical Interventions for Physiological Actuation

Alternatively to cognitive reminders, devices like "Dual-Blink" actively force eyelid closure through induced screen blurring, light mechanical touches, or periocular air micro-puffs . It bypasses the user's voluntary filter by biologically triggering the tear reflex and causing an instantaneous 36% increase in the blink rate. The level of intrusion is intolerable in normal operational workplaces and induces mild irritation, combined with the high fragility of the moving pneumatic parts [21] [15].

F. Wearable Egocentric Vision

Egocentric vision mounts a front-facing camera on the glasses, algorithmically processing exposure and tracking attention towards multiple screens. It offers exceptional semantic certainty, ignoring light pollution and functioning in any mobility context to measure the user's real exposure. Processing these videos, however, requires constant streaming to cloud servers, totally negating personal privacy, introducing latencies, and draining batteries in a few minutes [22] [23].

Current wearable systems frequently compromise on user ergonomics, operational efficiency, and data privacy due to their reliance on complex multi-sensor arrays and external cloud processing. The critical need to overcome these inherent trade-offs drives the introduction of a highly streamlined and minimal architecture: a single camera paired with integrated edge computing capabilities.

By analyzing the visual data stream entirely locally and in real time, this system circumvents the need for external data transmission, thereby guaranteeing the absolute user privacy lacking in previous models, while simultaneously eliminating unnecessary hardware overhead and latency. Furthermore, answering the demand for seamless daily integration, the device features silent haptic micro-vibrations specifically designed to enforce the 20-20-20 rule for eye strain relief. This creates an unobtrusive interaction model that directly responds to the necessity of perfectly balancing functional precision, physical invisibility, and long-term computational sustainability.

III. Methodology

This study proposes a wearable smart glasses system designed to mitigate Computer Vision Syndrome (CVS) by automating the 20-20-20 rule. The proposed system utilizes a micro-camera integrated into the frame of the glasses to capture the user's field of view, an embedded microprocessor for on-device inference, and a haptic motor to deliver physical prompts.

A. Dataset Acquisition and Setup

A significant challenge in developing robust on-device inference for wearable systems is the domain gap between publicly available image datasets and the actual first-person perspective (egocentric vision) of a user. Standard datasets predominantly feature third-person viewpoints, which fail to capture the unique angles, lens distortions, and glare dynamics of viewing light-emitting screens from a head-mounted micro-camera. To address this, a custom egocentric dataset was constructed.

Data was collected using a micro-camera module matching the field of view and technical specifications of the final smart glasses prototype. This camera was mounted on a wearable frame to naturally simulate human head movements, posture variations, and the specific viewing angles associated with continuous screen usage.

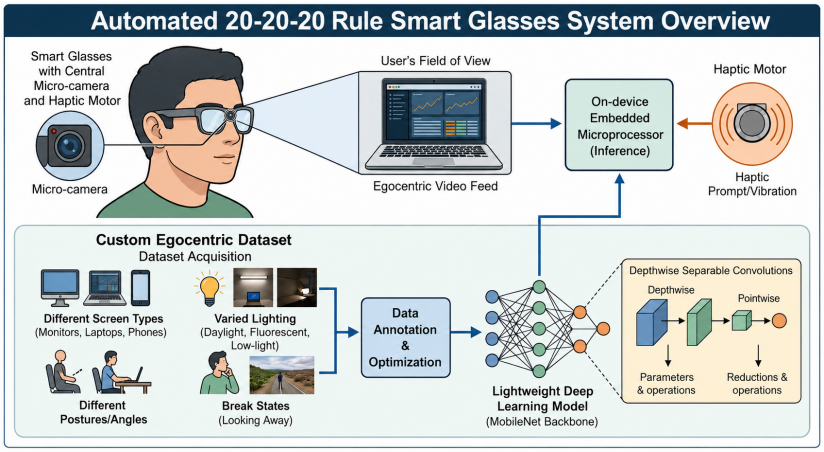


Fig. 2. Flowchart of the adaptive 20-20-20 rule operation logic and haptic feedback pipeline.

1) Collection Parameters: The dataset comprises approximately 982 annotated frames across two binary classes: *screen-active* (the user is looking at a screen) and *screen-absent* (the user is looking away, satisfying the rest condition of the 20-20-20 rule). The training set consists of 686 images, with 370 frames in the screen-active class and 316 in the screen-absent class.

To ensure robustness across real world environments, the data capture process encompassed the following scenarios:

- **Screen Diversity:** Several screen types were recorded, including desktop monitors, laptops, and smartphones, accounting for differences in aspect ratios, bezel sizes, and backlighting intensities.
- **Environmental Lighting:** Sessions were conducted across multiple ambient lighting conditions, including natural daylight and indoor artificial lighting. Expanding coverage to low-light and mixed-lighting environments is identified as a priority for future data collection.
- **Viewing Dynamics:** Footage was captured at natural viewing distances and head tilt angles characteristic of Computer Vision Syndrome (CVS) postures, including slight downward tilts for laptop use and near-horizontal gaze for desktop use.
- **Break States (Negative Samples):** The dataset includes footage of the user looking away from all screens toward distant objects, open space, and non-emissive surfaces to train the system to reliably recognise the rest condition.

2) Confounding Elements: A key challenge for binary screen-detection in real environments is the presence of visually similar stimuli that could generate false positives. Potential confounders include windows with high outdoor luminance, reflective or glossy surfaces, and televisions or projectors outside the user's primary working environment.

To address this, confounding samples - such as bright windows and reflective surfaces - are incorporated into the negative class, allowing the model to learn the discriminative features that distinguish an emissive screen from other high-luminance or reflective stimuli. This approach avoids the need for additional label categories while naturally extending the decision boundary of the classifier to handle real-world confounders.

B. Annotation and Optimisation

The collected visual data was manually annotated to categorise frames into the two target states. Inter-annotator consistency was ensured by a single annotator applying a fixed labelling protocol, minimising label noise given the dataset's scale. Finally, the dataset was optimised to accommodate the strict memory and computational constraints of the system's embedded microprocessor, including frame resolution reduction and format conversion suitable for the target inference engine.

C. Inference Model Architecture and Training

To accurately detect screen exposure in real-time on computationally limited hardware, the system employs a transfer learning approach built upon MobileNetV1 [24], a class of highly efficient convolutional neural networks designed specifically for mobile and embedded vision applications.

1) Architecture: Standard convolutional neural networks require substantial computational resources, making them unsuitable for continuous edge inference on wearable devices. MobileNetV1 overcomes this by replacing standard convolutions with depthwise separable convolutions, which factorize a standard convolution into two stages: a depthwise convolution that applies a single filter per input channel, followed by a pointwise convolution that linearly combines the outputs. This factorization drastically reduces both computational cost and model size, making real-time inference feasible on microcontrollers.

The specific variant employed is **MobileNetV1 96×96 α=0.1**, where the input resolution is 96×96 pixels (corresponding to 6,912 input features) and the width multiplier $\alpha = 0.1$ uniformly scales down the number of filters in each layer, yielding an extremely compact model suitable for the target embedded processor. The final dense layer of the pretrained backbone was removed and replaced with an output

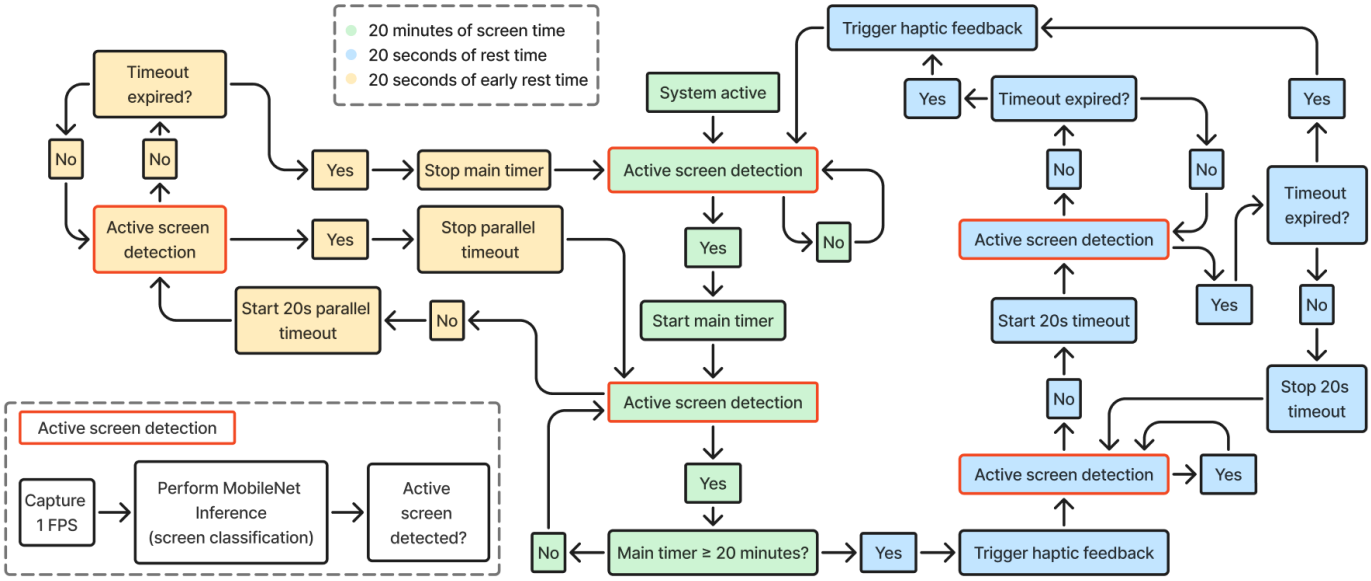


Fig. 3. Flowchart of the adaptive 20-20-20 rule operation logic and haptic feedback pipeline.

layer of two neurons corresponding to the binary classification task (*screen*, *no_screen*).

D. Operation Logic and Haptic Feedback

The inference pipeline is designed to monitor continuous screen usage. To preserve battery life, the camera samples the user's field of view at a fixed, low frame rate (e.g., 1 frame per second). The captured frames are fed into a MobileNet classifier, which outputs a probability score indicating the presence of an active computer screen.

The system maintains a rolling exposure timer. If the model consistently detects a screen in the user's field of view for 20 continuous minutes, the microprocessor triggers a haptic motor. This haptic feedback serves as a discreet physical cue, prompting the user to break their screen-staring state without relying on visual or auditory alerts that could be disruptive in office environments. Upon receiving this haptic cue, the user is advised to look away at an object approximately 20 feet away for 20 seconds.

Once the camera ceases to detect an active screen within the user's field of vision, it initiates a 20-second break countdown. Upon the timer's expiration, the system notifies the user with the haptic feedback signaling that the break is over. Following this, as soon as the system recognizes a screen in the user's field of view again, it restarts the 20-minute timer.

If, during the 20-minute "screen usage" window, the system stops detecting an active screen for at least 20 continuous seconds, the main 20-minute timer resets automatically. The system assumes that a sufficient break has already been taken naturally. In this scenario, no haptic feedback is provided to the user, ensuring that their workflow remains completely uninterrupted.

IV. Experimental Results and Discussion

A. Experimental Setup and Implementation Details

To achieve high accuracy with limited computational resources, a transfer learning approach was performed using the Edge Impulse platform for the core vision system. The raw image data was ingested and preprocessed to fit the constraints of edge computing. All input images were resized to a resolution of 96x96 pixels using a "fit shortest" scaling mode, resulting in an input layer of 27,648 features (accounting for the RGB color channels).

The base model selected was a pre-trained MobileNetV1 (configured for 96x96 inputs with an alpha multiplier of 0.1). The architecture was finalized with a custom dense layer consisting of 8 neurons and a dropout rate of 0.1 to mitigate overfitting, followed by a 2-node output layer for the binary classification task **(screen vs. noscreen)**. The network was trained over 40 epochs using a learning rate of 0.0005, leveraging GPU acceleration. Data augmentation techniques were enabled during the training phase to increase the dataset's variance and improve the model's generalization capabilities.

B. Performance Evaluation

To validate the effectiveness of the proposed detection system, we evaluated the model using standard classification metrics: accuracy, precision, recall, and F1-score. The evaluation was performed on two distinct datasets: a validation set **(N=138)** to tune the model, and a completely unseen hold-out test set **(N=156)** to verify its true generalization capabilities.

Furthermore, to assess the model's suitability for deployment on resource-constrained edge devices, we compared the unquantized **float32** model against an optimized **int8** quantized version. The aggregated results are presented in Table I.

TABLE I
PERFORMANCE COMPARISON BETWEEN INT8 AND FLOAT32 INFERENCE

Dataset	Type	Acc.	Prec.	Recall	F1	Loss
Validation (N=138)	int8	94.93%	0.95	0.95	0.95	0.17
Validation (N=138)	float32	95.65%	0.96	0.95	0.95	0.16
Test (N=156)	int8	94.87%	0.95	0.95	0.95	0.15
Test (N=156)	float32	95.51%	0.96	0.95	0.96	0.13

Note: Macro-averages are reported for Precision, Recall, and F1-Score to account for the two classes.

C. Discussion

The experimental results demonstrate the high reliability and robustness of the proposed method. On the unseen test set, both the float32 and the int8 models achieved a strong accuracy of 95.51% (float32) and 94.87% (int8). The macro-average F1-score remained stable at 0.95 across both data types, indicating that the system is equally capable of identifying both the presence and absence of a screen without bias toward a specific class.

A key finding from this analysis is the minimal impact of quantization on the model's predictive power. The int8 model exhibited negligible performance degradation compared to its float32 counterpart, while simultaneously dropping the computational load. Given the sub-second latency and the minimal memory footprint on an ESP32 architecture, this approach validates the feasibility of embedding complex computer vision tasks directly on-device, making it highly suitable for smart wearable architectures and cognitive assistance systems, such as smart glasses, where privacy, power efficiency, and real-time processing are paramount.

D. Edge Deployment and Hardware Optimization

A critical requirement of this project is the seamless deployment on microcontrollers. The trained model was compiled into a standard Arduino library utilizing the Edge Impulse EON™ Compiler, which successfully reduced RAM usage by 18% and Flash (ROM) usage by 34% without sacrificing accuracy.

Performance profiling was conducted targeting an Espressif ESP-EYE (ESP32 running at 240MHz). For the selected quantized int8 model, the estimated resource consumption is highly efficient: it requires only 194.9 KB of RAM and 182.1 KB of Flash memory. The total inference latency per image on this hardware is estimated at 934 ms (15 ms for digital signal processing and 919 ms for the neural network inference).

E. Design Prototype

To validate the physical feasibility of the proposed wearable system, a functional smart glasses prototype was designed and developed. The prototype was modeled in SolidWorks and engineered specifically around the dimensional constraints of the current ESP32 camera module used for edge inference. The frame geometry was therefore optimized to accommodate the embedded electronics.

The prototype was fabricated through additive manufacturing using 3D printing techniques, enabling rapid iteration and ergonomic evaluation of the wearable structure. In addition to the allocation for the ESP32 camera board, the architecture of the frame also integrates a dedicated housing for the haptic motor connected to the embedded processing unit. This configuration allows the system to deliver localized vibration feedback directly to the user during prolonged screen exposure events.

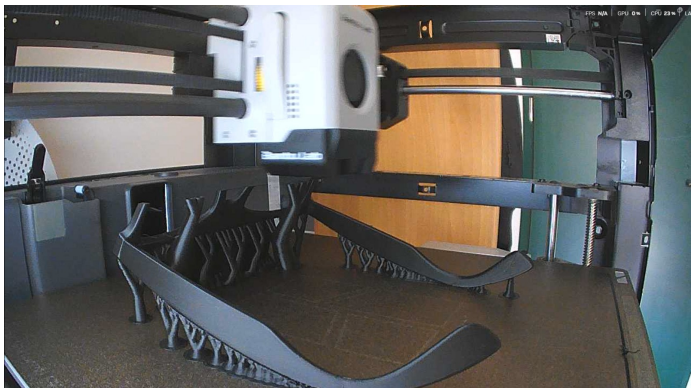


Fig. 4. 3D-Printed Wearable Prototype



Fig. 5. Prototype of the smart glasses system

The camera module was positioned frontally on the glasses frame in order to capture an egocentric point of view corresponding to the user's natural visual perspective while interacting with digital devices. This frontal placement enables continuous monitoring of the user's workspace and supports reliable real-time screen detection during everyday computer usage scenarios.

It is important to note that the current prototype presents dimensional limitations related to the off-the-shelf hardware components adopted during the experimental phase. Consequently, the prototype primarily serves as a proof-of-concept intended to demonstrate the designed interaction experience and the operational feasibility of the system. From an in-

dustrial design and engineering perspective, future product development would allow substantial miniaturization and optimization of the electronic architecture, enabling the complete integration of the sensing and feedback system within the standard dimensions of commercially viable smart glasses.

V. Conclusion

This work addresses the growing problem of Computer Vision Syndrome (CVS) caused by prolonged near-vision use of digital screens in occupational settings. To respond to this challenge, an adaptive assistive system in the form of smart glasses was developed, specifically designed to automate the 20-20-20 rule during computer use. Unlike traditional passive approaches, the proposed solution does not act as a simple static timer, but operates as a dynamic system that adapts in real time to user behavior, ensuring adequate visual rest through the continuous modulation of breaks according to the logic illustrated in the Fig. 3. This objective is achieved using strictly a forward-facing camera and an edge computing processor running an AI agent dedicated to local inference. Furthermore, the integration of a micro-vibration module allows for the delivery of effective physical feedback that does not excessively annoy the user, enabling them to avoid abruptly interrupting their workflow. Crucially, this streamlined approach achieves accurate environmental tracking and operationalizes the 20-20-20 rule without necessitating the complex, multi-modal sensor fusion architectures, such as Inertial Measurement Units (IMUs), spatial LiDAR, or localized physiological sensors, frequently relied upon in contemporary wearable computing literature. Future work will focus on three main directives to extend the robustness and applicability of the developed technology. First, we plan to test the prototype in a real-world context with an adequate sample of individuals in order to scientifically measure its tangible effectiveness on ocular well-being. Second, the system's detection capabilities will be broadened by expanding our custom database to include other pervasive digital screens, specifically training the model to recognize also smartphones and tablets. Finally, we aim to maximize the AI model's accuracy by significantly increasing the volume of reference images in our training dataset, thereby exposing the neural network to a wider variety of real world environments, reflections, and complex lighting conditions.

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